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**Cooling Design for a Wedge-Shaped LSC Panel with 19 LSC-Duct Modules**

## 1. Project Description

The project objective is to develop a liquid-cooled heat sink for a wedge-shaped Luminescent Solar Concentrator (LSC) / photovoltaic (PV) unit that maintains the PV temperature at or below 35°C under peak operating conditions while also satisfying the hydraulic and power constraints of the system. Each unit consists of a 0.01 m × 1 m PV strip coupled to a 1 m × 0.05 m wedge concentrator. The incident irradiance is 1000 W/m$^2$, the light concentration factor on the PV is 1.2, the PV absorptance is 0.90, and the electrical efficiency is 25% of absorbed radiation. The coolant enters at 27°C into the channel with a width of 0.01 m while the channel height is selected as part of the design below. In addition, the fluid velocity must remain below 5 m/s, and the pumping power for the full 1 m × 1 m panel must remain within the specified fraction of the electrical power generated. The coolant must also provide freezing protection for outdoor conditions near -10°C.

## 2. System Schematic

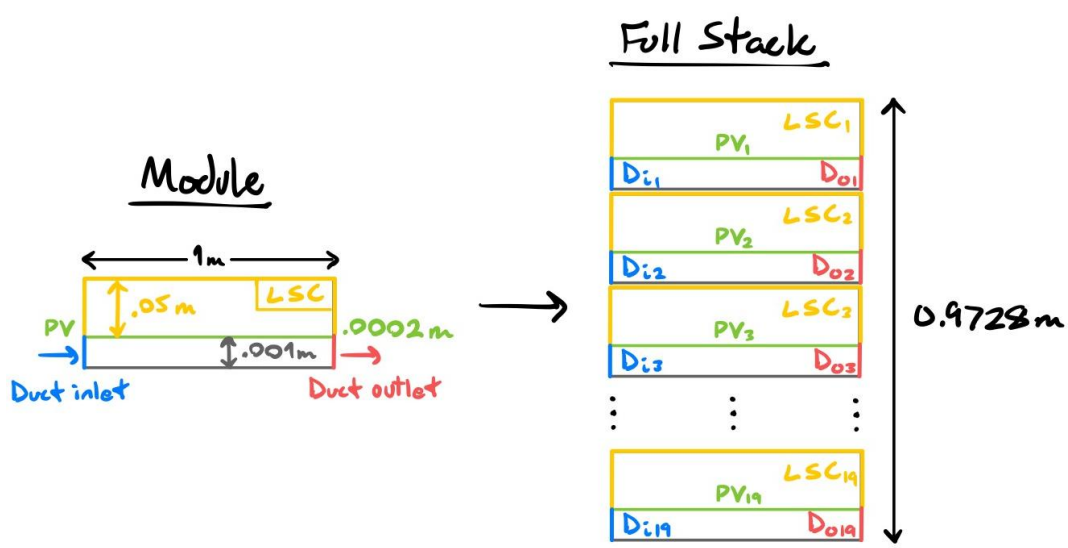


**Figure 2.1 – System Schematic**

## 3. Assumptions

1. The system operates at steady state under peak solar loading.

2. All absorbed solar energy not converted into electricity is transferred into the coolant.
3. The analysis is restricted to the concentrator, PV, and cooling channel unit; external piping and fitting losses are neglected.
4. The cooling channel has smooth walls and a rectangular cross section of fixed width .01 m and selected height H.
5. Coolant thermophysical properties are evaluated at the bulk mean fluid temperature.
6. Radiation and external convection losses from the modeled unit are neglected in the preliminary design stage.
7. The pump efficiency is initially assumed constant at 70% for first-pass calculations.
8. The propylene glycol/water mixture is selected to satisfy freezing protection near -10°C.
9. The PV thickness is treated as a fixed assumed input for the initial spreadsheet and will later be verified with a vendor datasheet.
10. The worst-case thermal design condition is used so that the resulting design is conservative.
11. The coolant flow is assumed to be fully developed laminar flow with a uniform heat flux. $N_u = 4.36$ .

## 4. Analysis

### 4.1 Symbolic Maximum PV Temperature

The maximum PV wall temperature, $T_{PVmax}$, can be calculated using the following equation:

$$T_{PVmax} = T_o + \frac{q''}{h} \quad [°C] \tag{1}$$

where $T_o$ is the duct outlet temperature in $°C$, $q''$ is the heat flux at the PV per duct in $\frac{W}{m^2}$, and $h$ is the heat transfer coefficient in $\frac{W}{m^2 * K}$.

To obtain $T_o$, Eq. 2 is utilized:

$$T_o = T_i + \frac{q_c}{\dot{m} * c_p} \quad [K] \tag{2}$$

where $T_i$ is the duct inlet temperature in $K$, $q_c$ is the heat rejected to the coolant in $W$, $\dot{m}$ is the coolant mass flow rate per channel in $\frac{kg}{s}$, and $c_p$ is the coolant specific heat in $\frac{J}{kg * K}$.

To obtain $q_c$, Eq. 3 is utilized:

$$q_C = P_{abs} - P_{elec} \quad [W] \tag{3}$$

where $P_{abs}$ is the power absorbed at the PV in $W$ and $P_{elec}$ is the electrical power from the PV in $W$.

To obtain these two values, Eq. 4-6 are used:

$$P_{abs} = G * C_f * A_{PV} * \alpha \quad [W] \tag{4}$$

$$P_{elec} = \eta_{PV} * P_{abs} \quad [W] \tag{5}$$

$$A_{PV} = W_{PV} * L_{PV} \quad [m^2] \tag{6}$$

where $G$ is the incident irradiance of the PV in $\frac{W}{m^2}$, $C_f$ is the concentration factor on the PV, $A_{PV}$ is the area of the PV in $m^2$, $\alpha$ is the PV absorptance, $\eta_{PV}$ is the PV electrical efficiency of the absorbed power, $W_{PV}$ is the width of the PV component in $m$, and $L_{PV}$ is the length of the PV component in $m$.

Finally, to obtain $q''$ in Eq. 1, the following equation is utilized:

$$q'' = \frac{q_c}{A_{PV}} \quad \left[\frac{W}{m^2}\right] \tag{7}$$

The equation for $h$ in Eq. 1 is determined in the following section, Section 4.2, as seen in Eq. 8.

### 4.2 Symbolic Heat Transfer Coefficient

In order to calculate the heat transfer coefficient, $h$, for Eq. 1, the following equation is used:

$$h = N_u * \frac{k}{D_h} \quad \left[\frac{W}{m^2 * K}\right] \tag{8}$$

where $N_u$ is the Nusselt Number for fully developed laminar flow with a uniform heat flux inside the duct, $k$ is the coolant thermal conductivity in $\frac{W}{m^2 * K}$, and $D_h$ is the hydraulic diameter of a rectangular duct in $m$.

The hydraulic diameter $D_h$ is calculated using Eq. 8:

$$D_h = \frac{2 * W * H}{W + H} \quad [m] \tag{9}$$

where $W$ is the channel width in $m$ and $H$ is the channel height in $m$.

### 4.3 Symbolic Produced Electric Power from a 1 m * 1 m Panel

In order to determine the electric power produced by a 1 m * 1 m panel, $P_{elec,panel}$, the following equation is used:

$$P_{elec,panel} = N * P_{elec} \quad [W] \tag{10}$$

where $N$ is the number of duct modules per 1 m panel and $P_{elec}$ is the electrical power from the PV in $W$, as calculated above in Eq. 5.

The number of duct modules is calculated using Eq. 11:

$$N = \frac{1}{H_{LSC} + H + t_{PV}} \tag{11}$$

where $H_{LSC}$ is the height of the LSC component in $m$ and $t_{PV}$ is the thickness of the PV component in $m$. This is then value is rounded down to the nearest whole number as there cannot be a fraction of a module.

### 4.4 Symbolic Pumping Power

For the necessary pumping power per panel, $P_{pump,panel}$, Eq. 12 is used:

$$P_{pump,panel} = N * P_{pump,duct} \quad [W] \tag{12}$$

where $P_{pump,duct}$ is the necessary pumping power per duct in $W$.

To find $P_{pump,duct}$, the following equation is used:

$$P_{pump,duct} = \frac{\Delta P * \dot{V}}{\eta_{pump}} \quad [W] \tag{13}$$

where $\Delta P$ is the pressure drop per duct in $Pa$, $\eta_{pump}$ is the pump efficiency, and $\dot{V}$ is the volumetric flow rate in $\frac{m^3}{s}$ which is calculated using Eq. 14:

$$\dot{V} = \frac{\dot{m}}{\rho} \quad \left[\frac{m^3}{s}\right] \tag{14}$$

where $\rho$ is the coolant density in $\frac{kg}{m^3}$.

To obtain $\Delta P$, Eq. 14 is used:

$$\Delta P = f * \frac{L_{PV}}{D_h} * \frac{\rho * u^2}{2} \quad [Pa] \tag{15}$$

where $f$ is the friction factor in the duct and $u$ is the mean duct velocity in $\frac{m}{s}$.

The friction factor is found using Eq. 15 and is calculated as follows:

$$f = \frac{64}{Re_{D_h}} \tag{16}$$

where $Re_{D_h}$ is the Reynold's number inside the duct which is obtained from Eq. 16 below:

$$Re_{D_h} = \frac{\rho * u^2 * D_h}{\mu} \tag{17}$$

where $\mu$ is the coolant viscosity.

In order to calculate the mean duct velocity used in Eq. 15 and 17, the following equation is required:

$$u = \frac{\dot{V}}{A_{CS}} \quad \left[\frac{m}{s}\right] \tag{18}$$

where $A_{CS}$ is the cross-sectional area of the channel which is given by Eq. 19:

$$A_{CS} = W_{channel} * H \quad [m^2] \tag{19}$$

where $W_{channel}$ is the width of the channel in $m$.

## 5. Design Process

### 5.1 Initial Starting Values for Mass Flow Rate, Channel 'H', & Coolant Mixture

To begin the design process, I initially selected values for the channel height, coolant flow rate, and the coolant mixture that were generally reasonable and consistent with the parameters of the project. Since the system has a wedge shaped luminescent solar concentrator, the design needed to balance thermal control with a compact repeated-unit geometry suitable for an integrated panel, as it was coupled to a PV strip and a cooling channel as well. Wedge-shaped LSCs are also appropriate for building integrated photovoltaic applications and have been shown to provide useful performance advantages in that configuration (Hughes et al.).

For the initial design, I selected a height of 0.004 m for the channel. I chose this since it is small enough to promote convection in a narrow rectangular channel, while still being large enough to avoid a small flow of passage. Since the channel wedge is fixed at 1 cm, the main geometric variable to adjust was the channel height to adjust thermal performance.

For the coolant, I used a propylene glycol/water mixture as suggested and used a 30 vol% PG mixture which provides protection below the required design minimum. Avoiding a concentrated mixture was necessary as increasing the concentration improves freeze resistance but increases viscosity and generally worsens heat transfer performance (DOWFROST Technical Data).

For initial flow rate, I used m = 0.0014 kg/s as a conservative starting value since it was large enough to make a meaningful convective cooling process while being low enough to expect the channel velocity and pumping power to remain comfortably below limits.

### 5.2 Initial $T_{PV}$ Calculation & Pumping Power Percentage

Using the starting values described above, I performed a full initial calculation for one duct module and then scaled the results to the full 1m x 1m panel.

| Parameter | Units | Value | Notes |
|---|---|---|---|
| Channel height (H) | m | 0.004 | Width of one duct module |
| PV thickness (tpv) | m | 0.0002 | Thickness of PV panel |
| Mass flow rate ($\dot{m}$) per channel | kg/s | 0.0014 | Mass flow rate of the coolant into the channel |
| Pump efficiency, h | - | 0.7 | Provide justification or datasheet for this value |
| Coolant type | **30 vol% DOWFROST / water** | | |
| PG mass fraction | - | 0.294 | Choose based on freezing protection from chart |
| Coolant density, ρ | kg/m³ | 1022.39 | At bulk mean temp (vendor data) |
| Coolant viscosity, μ | Pa·s | 0.002205 | At bulk mean temp (vendor data) |
| Coolant specific heat, cp | J/(kg·K) | 3886.6 | At bulk mean temp (vendor data) |
| Coolant thermal conductivity, κ | W/(m·K) | 0.45664 | At bulk mean temp (vendor data) |
| **GIVEN VALUES** | | | |
| LSC length | m | 1 | Wedge concentrator length |
| LSC height | m | 0.05 | Wedge concentrator height |
| Channel width | m | 0.01 | Width of channel |
| PV width | m | 0.01 | Width of PV panel |

| | | | |
|---|---|---|---|
| PV length/Channel length | m | 1 | Length of PV panel |
| Incident Irradiance, G | W/m² | 1000 | Peak design irradiance |
| Concentration factor on PV | - | 1.2 | |
| PV absorptance | - | 0.9 | Fraction of light absorbed |
| PV electrical efficiency (of absorbed) | - | 0.25 | Electric conversion fraction of absorbed |
| Coolant inlet temperature | °C | 27 | Design inlet temperature |
| Maximum PV/duct wall temperature (limit) | °C | 35 | Design criterion – do not exceed |
| Velocity limit | m/s | 5 | Erosion minimization |
| Pump power fraction limit | - | 0.05 | ≤ 5% of panel electric power |

**Table 5.1 – Initial Design Parameters**

By utilizing the equations described in Section 4, all but three values in Table 5.2 can be calculated. For the remaining values, the following equations will apply:

$$q_{panel} = N * q_c \quad [W] \tag{20}$$

$$P_{pump_{\text{limit}}} = 0.05 * P_{elec,panel} \quad [W] \tag{21}$$

$$Pump\ Fraction = \frac{P_{pump,panel}}{P_{elec,panel}} \tag{22}$$

where $q_{panel}$ is the total heat to remove from the panel in $W$, $P_{pump_{\text{limit}}}$ is the maximum power bound of the pump in $W$, and the $Pump\ Fraction$ is the total pump power over the total electric power of the panel.

| Result | Units | Value | Calculated in EQ |
|---|---|---|---|
| **A. Heat Load (per duct module)** | | | |
| PV area | m² | 0.01 | 6 |
| Absorbed power at PV | W | 10.8 | 4 |
| Electrical power from PV | W | 2.7 | 5 |
| Heat rejected to coolant (per duct) | W | 8.1 | 3 |

| | | | |
|---|---|---|---|
| Number of duct modules per 1 m panel | - | 18 | 11 |
| Panel total heat to remove | W | 145.8 | 20 |
| Panel total electric power | W | 48.6 | 10 |
| Pump power limit (≤ 5% of electric power) | W | 2.43 | 21 |
| **B. Flow & Thermal Sizing** | | | |
| Channel cross-section area (W × H) | $m^2$ | 0.00004 | 19 |
| Hydraulic diameter (rectangular duct) | m | 0.005714286 | 9 |
| Heat flux at PV (per duct) | $W/m^2$ | 810 | 7 |
| Outlet temperature (T_o) | K | 301.6386313 | 2 |
| Volume flow rate ($\dot{V}$) | m3/s | 1.36934E-06 | 14 |
| Mean velocity in duct (u) | m/s | 0.034233512 | 18 |
| Reynolds number (Re_Dh) | - | 90.70294785 | 17 |
| Nusselt number (assumption & source) | - | 4.36 | Assumption 11 |
| Heat transfer coefficient, h | $W/(m^2 \cdot K)$ | 348.41632 | 8 |
| Max PV wall temperature (T_PVmax) | °C | 30.81343626 | 1 |
| **C. Pressure Drop & Pumping** | | | |
| Friction factor (correlation used) | - | 0.7056 | 16 |
| Pressure drop per duct | Pa | 73.97519538 | 15 |
| Pump power per duct | W | 0.00014471 | 13 |
| Total pump power per panel | W | 0.002604786 | 12 |
| Total pump power/Total panel electric power (Pump Fraction) | - | 5.35964E-05 | 22 |

**Table 5.2 – Initial Design Calculated Values**

### 5.3 Design Iterations to Bring $T_{PV}$ & Pumping Power Percentage to Desired Values

After completing the initial calculation, I used the output values to understand how the design should be refined. We see that from the initial results with H = 0.004m and mass flow rate = 0.0014 kg/s already produced a low maximum PV temperature of 30.81 C and a

pumping power fraction of $5.36 * 10^{-5}$. This does satisfy the thermal and pumping power requirements, but it was more conservative than necessary. The cooling was stronger, and it reduced the number of LSC units that could fit within a 1m panel.

From these results I realized the design could be improved by having a more compact channel but still safely remain below the PV temperature limit. Since the goal requires maximization of the number of wedge units within the panel when having acceptable cooling. So logically I was to reduce the channel height. Lowering H decreases the unit pitch, allowing more LSC units to fit into the 1 m panel, and also tends to improve convection by reducing the hydraulic diameter. However, reducing H too much can increase pressure drop, so the flow rate also had to be reconsidered at the same time.

I reduced the channel height from 0.004m to 0.001m, then i also reduced the mass flow rate from 0.0014 kg/s to 0.0004 kg/s per channel. I changed this as the original flow rate was unnecessarily high to meet the temperature requirement. I understood that the flow rate could be lowered substantially without risking overheating. Reducing the mass flow rate also keeps the pumping power small and makes the design more efficient.

The coolant mixture was not changed during the iteration process. I kept the 30 vol% propylene glycol/water mixture because freeze protection below −10 °C remained a fixed requirement, and this mixture already met that need while avoiding a more concentrated glycol solution than necessary (DOWFROST Technical Data). Therefore, the iteration focused mainly on the geometric and flow variables rather than the coolant composition.

After updating the design, the final configuration gave a maximum PV temperature of **32.95 °C**, which is still below the required limit of **35 °C**. The final design also gave a total pumping power of **0.008868 W** and a pumping-power ratio of **0.0001729**, which is still far below the allowable limit. Compared to initially having a pumping power of **0.002604 W** and a ratio of **5.35964E-05.** Although the final PV temperature is slightly higher than the initial calculation, it remains safely within the acceptable range, and the revised design is more appropriate as we can use 19 LSC units per 1m panel instead of 19 in the initial case. This means the final design better satisfies the project goal of maintaining thermal performance while having maximized panel packing efficiency.

| **Final Design Parameters** | **Magnitude** |
|---|---|
| $T_{PV,\,max}$ (in °C) corresponding to this set of parameters | 32.95 |
| Number of LSC units per 1 m | 19 |
| Mass flow rate per channel (kg/s) | 0.0004 |
| Total mass flow rate for 1m x1m panel (kg/s) | 0.0076 |
| Channel H (m) | 0.001 |
| Pump efficiency | 0.7 |
| Total pumping power (W) | 0.008868 |
| Total electrical power generated (W) | 51.3 |

| | |
|---|---|
| Total pumping power/ Total electrical power generated | 0.0001729 |
| **Assumed Parameters** | **Comments** |
| PV thickness | The PV panel thickness is assumed to be 0.0002 m. |
| Channel Wall thickness | The channel wall thickness is assumed to be 0.0005 m. |

**Table 5.3 – Final Design Parameters**

## 6. CAD File

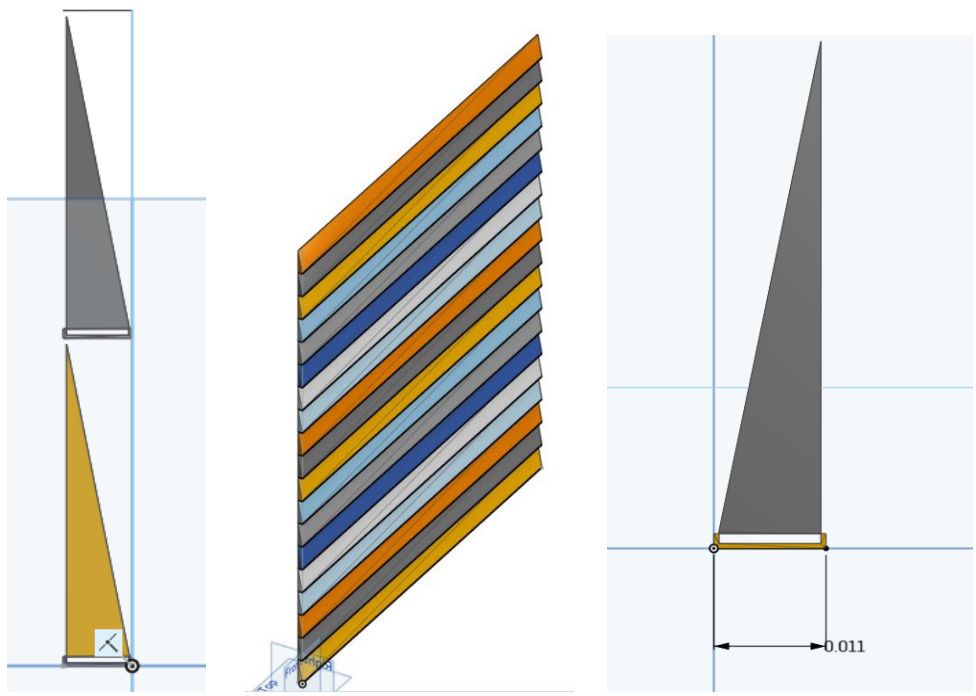


**Figure 6.1 – CAD Model**

## 7. Standards to Consider

Multiple standards and practices must be considered in the design of the LSC system. Since the coolant is propylene glycol/water mixture, the first consideration is safe handling, storage and use of the heat transfer fluid. OSHA's Hazard Communication Standard also requires that chemical hazards be communicated through labels and safety data sheets, so any real implementation of this design would need to ensure that the coolant is properly labeled (OSHA; DOWFROST Technical Data).

Another is Materials compatibility and corrosion control. Since the coolant flows in a close loop and is in constant contact with the walls, pumps, the materials must be compatible with the fluid. Otherwise, corrosion, or long-term degradation might occur. So,

the design should assume corrosion-resistant materials (DOWCAL 200 Heat Transfer Fluid; ASHRAE White Paper).

Coolant concentration is another design issue. The percentage of glycol cannot be selected only for freeze protection as increasing glycol concentration also raises the viscosity and reduces the thermal efficiency and performance. So, a moderate concentration such as the selected 30 vol% mixture is appropriate for the design (ASHRAE Building Readiness; DOWFROST Technical Data).

Disposal and environmental release should also be considered. Even though propylene glycol is usually preferred over the ethylene glycol concentrate where lower toxicity is desired. If we are talking from a design standpoint, the system should minimize leakage and allow for controlled draining systems (EPA Nonpoint Source Guidance).

## 8. Conclusion

This project developed a liquid-cooled heat sink design for a wedge-shaped luminescent solar concentrator and photovoltaic panel system. The purpose was to design this system with a temperature of or below 35 C while it is at peak operating conditions. At the same time, it should also satisfy hydraulic and pumping power constraints. Using the selected geometry, coolant properties, and flow assumptions, the final design was able to meet all the specifications.

The final design had a channel height of **0.001m,** a mass flow rate of **0.0004 kg/s per channel** and a 3**0 vol% propylene glycol/water mixture** for freeze protection. We were able to achieve a maximum of PV temperature of 32.95 C, which is safely below the design limit. The panel configuration was finally achieved with 19 LSC units per 1m panel. In addition to this we achieved a total pumping power for the 1m x 1m panel at **0.008868 W**, this gives a ratio of **0.0001729** relative to the electrical power produced. This shows that the thermal goal was achieved and it was without having a significant hydraulic penalty.

The design process showed the importance of having a balanced cooling performance with compact geometry. We initially designed a system where PV temperature was lower than necessary, but it used a larger channel height, and we were not able to have many LSC modules in the panel. So, if we reduce the heigh and lower the flow rate then we can add additional panels which then helps with adding more power.

Overall, our final design is a practical solution for cooling the LSC_PV system. We have met the thermal requirements, power limit, coolant velocity and adequate freeze protection for an operation outdoors. Therefore, our final design is

an acceptable liquid-cooling solution for the specified 1m x 1m wedge shaped LSC panel.

## 9. References


[1] ASHRAE, *ASHRAE Building Readiness*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, 2021.

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[3] DOW Chemical Company, *DOWCAL™ 200 Heat Transfer Fluid*, Dow Chemical Company.

[4] DOW Chemical Company, *DOWFROST™ Propylene Glycol-Based Heat Transfer Fluid: Product Information and Engineering Data*, technical data bulletin, Dow Chemical Company.

[5] DOW Chemical Company, *DOWFROST™ Heat Transfer Fluid*, Dow Chemical Company.

[6] M. D. Hughes, D. E. Smith, and D.-A. Borca-Tasciuc, "Performance of wedge-shaped luminescent solar concentrators employing phosphor films and annual energy estimation case studies," *Renewable Energy*, vol. 160, pp. 513–525, 2020.

[7] Occupational Safety and Health Administration, *Hazard Communication Safety Data Sheets QuickCard*, U.S. Department of Labor, 2024.

[8] U.S. Environmental Protection Agency, *National Management Measures to Control Nonpoint Pollution from Urban Areas*, EPA.


## 10. Appendix

Link to Excel Sheet - https://rpi.box.com/s/kn04gadq1uath8m6q1vjeuwbnij4zu6o

Link to CAD with 19 Wedges Stacked - https://cad.onshape.com/documents/035c1f0346f76740d1328417/w/a0d34a9de41c814209be3484/e/c9747f3c20f11d6ba05da76a?renderMode=0&uiState=69ec300ba99468cca116d190

Link to CAD with 1 Wedge - https://cad.onshape.com/documents/3cc4a1233aed339af3bbd351/w/b521631e71b6447065272059/e/35c7e8cd8e1f734d1a1d562c?renderMode=0&uiState=69ec367cd87e5caafe58ccd7

## 12. System-Level Considerations

A challenge to achieving very high optical efficiency in a real-world environment is precision and tolerance. A slight deviation from required specifications can have a large impact the longer the distance the light has to travel. A way to work around this is to increase the area of the reflectors and receivers. However, increasing the area directly impacts the heat transfer, and a larger area dissipates more heat meaning more heat is lost and the LSC is less efficient.